

Kurmathi Sai Prakash Reddy

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SUMMARY

Software Engineer with 3+ years of experience building secure, scalable, and cloud-native applications using React, .NET Core, Node.js, SQL Server and AWS. Strong expertise in backend API development, microservices, and database-driven systems, with hands-on experience delivering production software in insurance, analytics, and hospitality domains. Solid academic and practical foundation in Machine Learning, Applied AI and LLMs, including integrating CNN-based models into web applications. Passionate about clean code, system reliability, and building user-centric products.

WORK EXPERIENCE

Software Engineer, Marsh & McLennan

Aug 2024 – Present

- Contributed to the development of a cloud-native Risk Analytics and Insurance Claims Optimization platform supporting real-time workflows with high availability.
- Designed and implemented backend microservices using C#, ASP.NET Core Web API, and SQL Server for policy validation, claims processing, and real-time risk scoring.
- Integrated Apache Kafka for asynchronous messaging and event-driven communication between distributed services.
- Implemented secure data access software development using IAM, encryption at rest, and role-based authorization to meet HIPAA and SOX compliance requirements.
- Built and maintained CI/CD pipelines using Docker, AWS CodePipeline and CodeBuild, reducing deployment time by approximately 35%.
- Managed cloud infrastructure using AWS ECS, Lambda, and CloudFormation, enabling reliable and repeatable deployments.
- Implemented monitoring, logging, and alerting using CloudWatch dashboards and alarms to proactively detect system issues.
- Developed unit and integration tests using xUnit and Moq, achieving approximately 85% automated test coverage.

Software Engineer, NCR Corporation (NCR Voyix)

Jan 2022 – Jul 2023

- Contributed to the modernization of the Aloha Cloud hospitality platform by delivering full-stack solutions across UI, API, and database layers.
- Migrated legacy Ext JS and .NET systems to modern ReactJS, and .NET Core services, improving application performance and scalability by nearly 50%.
- Developed reusable React components to improve UI consistency and reduce development effort across multiple features.
- Designed and implemented API-first backend services to support third-party integrations and future product expansion.
- Optimized SQL Server queries and LINQ-based data access, reducing data retrieval latency by approximately 50%.
- Followed TDD and Agile methodologies, actively participating in sprint planning, code reviews, Root cause analysis, and production releases.

- Web Development Intern, The Sparks Foundation

Nov 2020 – Dec 2020

 - Developed a responsive web banking application using JavaScript, HTML, and CSS with secure fund transfer and balance inquiry features.
 - Deployed the application using GitHub Pages, achieving over 100 views within the first month.
 - Managed source control using Git, maintaining clean commit history and version control best practices.

MACHINE LEARNING & AI PROJECTS

- **Emotion-Based Content Recommendation Web Application (CNN, FER2013)**
Designed and implemented an end-to-end web application that detects real-time facial emotions using a pre-trained CNN fine-tuned on the FER2013 dataset. Integrated camera-based image capture for facial emotion analysis and dynamically redirected users to personalized Spotify and YouTube content based on detected emotions. Documented model training, fine-tuning strategy, system architecture, and inference workflow to demonstrate applied ML-to-product integration.
- **Multi-Agent AI Bedtime Story Generator (LLMs, GPT-3.5)**
Designed and implemented a multi-agent AI system to generate age-appropriate bedtime stories for children aged 5–10. Orchestrated multiple autonomous agents with distinct roles (story planning, narrative generation, feedback evaluation, and refinement) using GPT-3.5-turbo. Built prompt pipelines and agent coordination logic to ensure coherent storytelling and safety-filtered output, demonstrating applied LLM system design, prompt engineering, and AI-driven content generation.

EDUCATION

- 2023 – 2025

Master of Science in Computer Science, **University of Central Missouri**
- 2018 – 2022

Bachelor of Technology in Computer Science, **KL University, India**
Specialization in Data Science and AI

SKILLS

Languages	C#, C++, JavaScript, SQL, Java, Python
Frontend	ReactJS, Redux, HTML, CSS, Responsive Design
Backend	ASP.NET Core Web API, Node.js, REST APIs, Microservices, API Versioning, GraphQL, Distributed Systems
Databases	SQL Server, Amazon RDS, DynamoDB, Relational & NoSQL Design, Query Optimization
Cloud	AWS (ECS, Lambda, RDS, DynamoDB, CloudFormation, CloudWatch), IAM, CI/CD
DevOps	Infrastructure as Code, Automated Testing, Monitoring & Logging, Docker
Machine Learning & AI	Machine Learning, Deep Learning, CNNs, Computer Vision, Large Language Models (LLMs), Multi-Agent Systems, Prompt Engineering, Model Fine-Tuning, Model Inference Integration, NumPy, Pandas, ONNX Runtime
Practices	Agile/Scrum, TDD, Clean Code, Code Reviews, Root cause analysis, Documentation

CERTIFICATIONS

- AWS SysOps Associate Certification Training – Simplilearn
- Artificial Intelligence Foundations – NASSCOM
- DevOps & Cloud Computing – Bytexl
- Python for Data Science and AI – IBM (Coursera)
- Angular Essential Training – LinkedIn Learning